Group Project Brief

Real-World Database Prototype

Release: Week 7 | **Milestone:** Week 9 tutorial | **Project submission:** Week 11 (25 Oct 23:59) | **Poster showcase:** Week 12 or 13

1. Project Overview

In this group project, you will identify a real-world data management problem and design, implement and showcase a database-driven prototype that addresses it. The project integrates database knowledge developed through Week 10. The database solution is the primary technical focus, while the prototype and showcase communicate how that solution works in a realistic setting.

The assessed technical scope is limited to material taught by the end of Week 10. Material first introduced after Week 10 is not required.

2. Design Brief: Solve a Real-World Problem with a DBMS

Choose a real-world situation in which data must be stored, related, queried, updated and kept consistent. Design a relational database solution and a functional prototype that demonstrates how the database supports realistic users and tasks. Suitable domains include healthcare, education, sustainability, transport, retail, bookings, inventory, community services, research, facilities and resource management.

The problem should be substantial enough to require multiple related entities, meaningful business rules, non-trivial SQL queries, integrity mechanisms and at least one realistic multi-step transaction.

3. Group Formation

- Work in groups of up to 3 students.
- All members should contribute meaningfully to design, implementation, testing, documentation and preparation for the showcase.
- Nominate a project coordinator to organise tasks and communication.
- A smaller group does not automatically receive a reduced project scope.

4. Core Project Requirements

4.1 Problem and Requirements

- Define the real-world problem, intended users/stakeholders and value of the proposed database solution.
- Identify the data to be managed, important business rules and realistic user operations.
- Make requirements specific enough to guide and justify the database design.

4.2 Conceptual Design (ER Model)

- Create an ER model using the lecture notation with appropriate entities, attributes, relationships, cardinalities and participation constraints.

- Explain important modelling decisions and assumptions.
- Ensure the model is traceable to the stated requirements.

4.3 Logical Design and Normalisation

- Translate the conceptual model into a relational schema (i.e., RM diagram).
- Identify primary keys, candidate/unique keys, foreign keys and relevant constraints.
- Identify important functional dependencies and analyse/refine the schema using normalisation concepts covered by Week 10.
- Justify important design trade-offs.

4.4 SQL, Integrity and Transactions

- Implement the database in PostgreSQL and populate it with sufficient realistic or synthetic data.
- Provide meaningful SQL operations that support genuine requirements and appropriately use techniques covered in the unit.
- Implement suitable key, domain, referential and business-rule integrity constraints.
- Identify and demonstrate at least one realistic multi-step transaction. Explain transaction boundaries, failure/rollback behaviour and relevant concurrency considerations at a level consistent with material taught by Week 10.

4.5 Functional Prototype

Create a working prototype that connects to PostgreSQL and demonstrates the main features of your database solution. You may use Python or Java and may build a simple web, desktop or command-line interface. Interface complexity is not the goal; the prototype should make the database functionality clear, convincing and easy to demonstrate.

As part of the Week 11 submission, your group must submit a maximum 3-minute project showcase video demonstrating the prototype and its key features. Be creative in how you communicate the problem, solution and user experience. The most creative project showcase video will receive a Certificate of Appreciation from the COMP9120 teaching team. Creativity is encouraged, but the video must still clearly demonstrate the submitted prototype and database functionality.

4.6 Final Demonstration and Poster Showcase

After submission, groups will participate in a poster showcase event organised by the teaching team in Week 12 or Week 13. Each group must create a one-page poster, submit it with the Week 11 project package, and later print and display it at the showcase. During the event, groups will demonstrate the prototype, explain key design decisions, discuss their database solution with attendees and answer questions from the teaching team. The final demonstration is worth 15% of the project mark.

5. Week 9 Milestone - Proposal and Tutor Feedback

During the Week 9 tutorial, each group must show its proposal to the tutor and obtain formative feedback. Bring:

- a concise problem/challenge statement and intended users;
- key data requirements and business rules;
- a draft ER diagram;
- a preliminary relational schema or mapping plan;

- 3-5 representative operations/queries;
- a brief prototype plan and division of work;
- specific questions or design decisions for tutor feedback.

Record the main feedback received and use it to refine the project. This milestone is formative rather than a separate graded submission.

6. Week-by-Week Roadmap

Week	Focus	Recommended activities	Checkpoint/output
Week 7 - Release	Problem framing and setup	Form group; select problem; identify users, data and business rules; set up PostgreSQL/repository; plan prototype.	Agreed problem, roles and initial requirements.
Week 8	Conceptual/logical design	Develop ER model; map to relations; identify keys, constraints and functional dependencies; normalise; begin database implementation.	Draft ERD, relational schema and normalisation notes.
Week 9 - Milestone	Tutor feedback and transactions	Show proposal to tutor; obtain feedback; revise design; identify transaction scenario; continue SQL, integrity and prototype development.	Tutor-reviewed proposal and revised design.
Week 10	Integration and testing	Complete SQL, integrity and transaction behaviour; integrate prototype; test realistic workflows; draft report sections #1-#4; plan video and poster.	Feature-complete prototype, draft report, video/poster plan.
Week 11 - Submit	Final submission	Final QA; complete contribution summary and report; record/edit max 3-minute showcase video; finalise one-page poster; submit package.	Complete submission package.
Week 12 or 13	Poster showcase and final demonstration	Print the submitted poster; present the project; demonstrate the prototype; explain design decisions and answer questions.	Assessed final demonstration and poster showcase.

7. Submission Package - Week 11

Submit the following four items as a group:

1. Group contribution summary

Use the template provided by the teaching team. Clearly outline the contribution of each member of the group.

2. Project report

Structure the report explicitly around marking criteria #1-#4 so markers can readily locate the evidence used for assessment. Use the following four main sections: (1) Problem and Requirements; (2) Conceptual Design (ER Model); (3) Logical Design and Normalisation; and (4) SQL, Integrity and Transactions. Within each section, include the relevant rationale, evidence, diagrams, SQL/examples, testing and discussion. Supporting appendices may be used where appropriate.

3. Prototype showcase video - maximum 3 minutes

Show the main prototype features and how the database solution addresses the real-world problem. The video should be concise, engaging and creative. The most creative project showcase video will receive a Certificate of Appreciation from the teaching team.

4. One-page project poster

Submit a one-page poster that communicates the problem, solution, database design and prototype clearly. The group will later print this same poster and use it during the Week 12 or Week 13 poster showcase event organised by the teaching team.

8. Marking Distribution

The project is marked out of 100. Criteria #1-#4 are evidenced primarily through the report and technical submission; the functional prototype is evidenced through the working system and 3-minute showcase video; and the final demonstration is assessed at the Week 12/13 poster showcase.



Marking criterion	Weight
1. Problem and Requirements	10%
2. Conceptual Design (ER Model)	20%
3. Logical Design and Normalisation	20%
4. SQL, Integrity and Transactions	20%
5. Functional Prototype	15%
6. Final Demonstration	15%

10. Use of Generative AI

Generative AI is allowed for this open assessment, subject to the unit and University requirements. Any use must be acknowledged. Your group remains responsible for the correctness of the database design, SQL, code and documentation, and every member should be able to explain the submitted work.